

Ashutosh Naik

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EDUCATION:

M.S in Computer Science

University of Colorado Boulder, Boulder, Colorado

Expected May 2024

B.Tech. in Computer Engineering, GPA: 9.77/10

Sardar Patel Institute of Technology, Mumbai, Maharashtra

May 2022

Relevant Coursework: Object Oriented Programming, Artificial Intelligence and Soft Computing, Distributed Systems, Design and Analysis of Algorithms, Database Management Systems, Advanced Data Structures.

TECHNICAL SKILLS:

Programming Languages: Python | C++ | C | SQL.

Libraries and Frameworks: MPI | pandas | matplotlib | TensorFlow | PyTorch | NumPy | Flutter.

EXPERIENCE:

ML Intern, Skinzy Software Solutions

December 2020 – December 2021

- Developed an object detection and segmentation model to better identify and distinguish across more than 15 skin diseases.
- Fine-tuned neural network models such as MRCNN, UNET and utilized detectron2 library on custom datasets for segmenting skin diseases. Acquired an accuracy of over 90% across each class.
- Attained an accuracy of over 80% on jaundice detection task.
- Conducted a session and was in-charge of hiring and training new interns. Also collaborated with the App Dev team to identify major payment related bug in application.

Part-time Research Intern, Sardar Patel Institute of Technology

August 2020 – November 2020

- Collaborated on a project involving detection of cross events in Tweets. When an event occurs, it is widely discussed on Twitter. Other events related to primary one also emerges. These are cross events.
- Applied the BERT model to convert tweets into encoding vectors. These encodings were then used to measure similarity between tweets and for clustering.

INDEPENDENT PROJECTS:

Segmentation of Nuclei using Transformer based Architecture

January 2022 – April 2022

- Fine-tuned the TransUnet model on 4 nuclei segmentation datasets obtained under different laboratory conditions.
- Obtained a 64% higher IOU score than CNN based segmentation under similar training conditions.

Vigila – Application for General Safety

January 2021 - April 2021

- Designed and developed prototype for a community driven safety application in Flutter while working in a team of 3.
- Conducted a survey of over 375 people to evaluate public opinion of such application.
- Accepted at IEEE 2021 International Conference on Advances in Computing, Communication and Control (ICAC3'21) 7th Edition conference.

Neuro-Evolution Snake

June 2020

- Implemented classic snake game in python using pygame library.
- Implemented a genetic algorithm - a neural network using concept of neuro-evolution to 'see' in 8 directions.
- After a few generations network acquired ability to play, steering to avoid snake from hitting itself or surrounding walls and was able to score points by eating food.

Sudoku Solver

May 2020

- Designed a program that solves a sudoku problem from an image.
- Employed OpenCV library and a neural network to extract grid and identify numbers from an image.
- Utilized backtracking algorithm to solve actual sudoku grid.

CERTIFICATIONS:

- "Natural Language Processing in TensorFlow", Coursera, June 2020.
- "Convolutional Neural Networks in TensorFlow", Coursera, May 2020.
- "Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning", Coursera, May 2020.
- "Deep Learning Specialization", Coursera, May 2020.